PORTER: A Plan-Committed Router for Multi-Core Distributed Quantum Computers

***Abstract*—We present PORTER (*Plan-Oriented Router for Teleportation-Enabled aRchitectures*), a router for multi-core distributed quantum computers that minimises inter-core EPR-pair consumption. PORTER unifies the three inter-core communication primitives used in prior work — state teleportation (TELEDATA), gate teleportation (TELEGATE), and multi-gate cat-entangler packets (TELEGATEPACKET) — under a single *plan-committed* decision loop: where TELESABRE and DSABRE score every inter-core candidate together with its staging SWAPs in one flat action pool, PORTER first scores whole remote plans on their physical and DAG-lookahead merits alone, commits to the best one, and only then generates the preparation SWAPs specific to that plan, so preparation effort is never spent on a plan that is not selected. A two-tier capacity-relief mechanism proactively routes qubits toward reservoir cores when candidate plans are blocked by insufficient free capacity, and a deterministic, seed-free *temporal locality-greedy* initial layout co-locates qubits by temporally discounted interaction weight in first-use order. On MQT-Bench suites at 25, 64, and 80 logical qubits, a *single* PORTER run from this single layout posts the lowest geometric-mean EPR count of the three routers on every suite – reducing EPR against TELESABRE by 54.3%, 49.8%, and 65.8%, and beating DSABRE's best-of-three protocol (43.1 vs. 45.7, 86.4 vs. 95.5, and 14.0 vs. 21.0) – while routing 58–92× faster per attempt than DSABRE. Routed from identical externally supplied layouts across thirty instances, PORTER also matches or beats DSABRE's routing stage (fifteen wins to twelve, three ties). PORTER is the only router of the three to complete every benchmark: on an 80-qubit, 23,381-CX circuit it finishes in 37 s where both baselines exceed a uniform 900 s compiler timeout. The implementation, an independent replay verifier, and the full comparison harness are publicly available.**

***Index Terms*—distributed quantum computing, circuit routing, qubit teleportation, plan-committed routing, EPR minimisation, qubit mapping, lookahead heuristic**

I. INTRODUCTION

Distributed quantum computers (DQCs) interconnect several small, high-fidelity quantum processing units (QPUs, or *cores*) with inter-core links that support remote entanglement generation [1], offering a near-term path to scale beyond what a single monolithic chip can host. Compiling a circuit onto a DQC decomposes into three coupled sub-problems [2]: *(i)* statically assigning logical qubits to cores, *(ii)* choosing, for each non-local two-qubit gate, which communication primitive realises it, and *(iii)* routing and scheduling the resulting sequence of intra-core SWAPs and inter-core operations subject to communication-port capacity. Every communication primitive consumes one or more EPR pairs, and EPR generation is slow, noisy, and rate-limited on current hardware [3], [4], making EPR minimisation the dominant compilation objective.

PORTER — a *Plan-Oriented Router for Teleportation-Enabled aRchitectures* — addresses stage (iii), ordering intra-core SWAPs and inter-core operations to minimise EPR consumption, and ships a deliberately lightweight answer to stage (i): a deterministic, seed-free initial layout built in microseconds, so that the full pipeline needs neither an external partitioner nor a per-circuit layout search (external layouts remain supported, and are used throughout the evaluation as controlled probes). The name is also a job description: like a hotel porter, the router's task is to decide which guests (logical qubits) to carry to which rooms (cores), when, and by which means — and to do so only after a plan for the whole errand has been settled. Unlike DSABRE [5], which restricts itself to state teleportation (TELEDATA), PORTER follows TELESABRE [6] in also supporting gate teleportation (TELEGATE) via the cat-entangler protocol [7], [8], and extends it with a same-control *packet* primitive (TELEGATEPACKET) that amortises one EPR pair over an ordered run of consecutive remote CX gates sharing a control qubit, in the spirit of gate-sequence amortisation used by AutoComm [9].

The router departs from TELESABRE and DSABRE in *when* it commits to core-preparation SWAPs. Both score inter-core candidates and their staging SWAPs together in one flat action pool. PORTER instead separates the decision into two phases: first, every legal remote *plan* — a teleport direction, a single telegate, a same-control packet, or a capacity-blocked variant of any of these — is scored using only its physical and DAG-lookahead properties, independent of how many preparation SWAPs it would take to make it physically legal right now; only *after* one plan is selected does the router generate and apply preparation SWAPs specific to that plan, repeating until the plan becomes physically ready or no further preparation SWAP improves it. This plan-first ordering means preparation effort is never spent scoring or partially staging a remote action that a better one will preempt.

Contributions.

- **A plan-first remote-action loop** that separates candidate scoring from preparation-SWAP generation (Section III-D), unifying teleport, telegate, and same-control packet primitives under one direct/latent/lookahead gain decomposition (Section III-E).
- **A two-tier capacity-relief mechanism**: a breadth-first search for the nearest reservoir core when a plan is capacity-blocked (Section III-H), plus an opportunistic comparator that lets a severely blocked candidate preempt a merely-good normal one. An ablation (Section IV-F) shows this mechanism is what makes the deterministic-greedy design complete at scale: without it, four of the six 80-qubit benchmarks deadlock.
- **A deterministic, seed-free initial layout** — temporal locality-greedy (Section III-A): each CX pair is weighted by temporally discounted interaction multiplicity, qubits



Fig. 1. (a) A five-CX circuit on four logical qubits and (b) its dependency DAG (edges labelled by the shared qubit). The DAG front — the gates with no unexecuted predecessor — is shaded: initially $\{g_1, g_2\}$; once both execute it becomes $\{g_3\}$, then $\{g_4, g_5\}$.

are visited in first-use order and co-located with their heaviest placed neighbour under a per-core free-slot reservation. A single layout built in microseconds replaces per-router layout search entirely, yet outperforms DSABRE's SabreLayout-based best-of-three protocol end-to-end on every suite we test.

- **An independent replay verifier** (Appendix D) that re-executes the emitted operation stream against its own physical-layout simulation and checks gate coverage, DAG-order faithfulness, and every physical legality constraint (coupling-edge adjacency, port occupancy, cat-section isolation), used to certify correctness across 60 randomized circuits and the full evaluation suite — and itself validated by fault injection: ten classes of single-point stream corruption, every mutant rejected by the intended check.
- **A head-to-head evaluation** against TELESABRE and DSABRE on identical physical topologies at 25, 64, and 80 logical qubits (Section IV), under each router's own initial-layout protocol and under shared fixed layouts (with an external `pytket-dqc`-generated layout source in Appendix B).

The PORTER implementation, the replay verifier, the full cross-router comparison harness, and the raw per-circuit results are publicly available at https://github.com/ebony72/porter.

II. Background

A. Quantum Circuits and Routing

A quantum circuit acts on n logical qubits through a sequence of gates; we work in the restricted basis $\{\texttt{id}, \texttt{rz}, \texttt{sx}, \texttt{x}, \texttt{cx}\}$ used by MQT-Bench's Qiskit-opt3-transpiled circuits, i.e. arbitrary one-qubit (1Q) gates plus the two-qubit (2Q) CX gate. A circuit is represented as a directed acyclic graph (DAG): for every logical qubit, consecutive gates acting on it are connected by a dependency edge. The *front* (or *DAG front*) is the set of gates with no unexecuted predecessor; only front gates can execute or be routed towards execution at a given step. Figure 1 shows a five-gate circuit and its DAG; note that gates sharing a qubit (e.g. two CX gates with the same control) are chained by a dependency edge and therefore *never* appear in the front simultaneously — a fact that matters for packet collection (Section III-F).

Physical qubits are partitioned into *cores*; each core has an intra-core coupling graph, and pairs of cores may be connected by an inter-core link whose two endpoints are designated *communication ports* (Figure 2). A *layout* maps every logical qubit to a physical position. A 2Q gate is *local* if both operands share a core and *remote* otherwise. Routing a local gate that is not yet adjacent requires intra-core SWAPs; routing a remote gate requires an inter-core communication primitive.

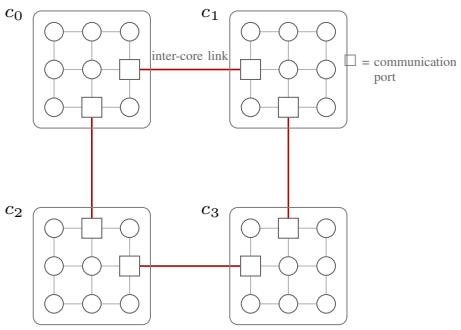


Fig. 2. Architecture A: a 2×2 array of cores, each a 3×3 grid of physical qubits (circles) with nearest-neighbour intra-core coupling (grey edges). Squares are communication ports; every inter-core link joins one port in each of two adjacent cores and is the only place EPR pairs are consumed. Architecture H (2×3 array of 4×4 cores, Section IV) follows the same pattern.

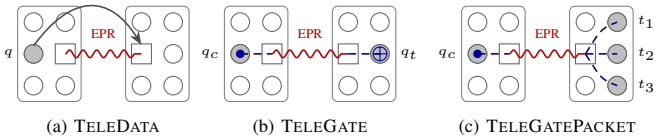


Fig. 3. The three communication primitives; each consumes one EPR pair between the two communication ports (squares). (a) TELEDATA moves qubit q (which must sit adjacent to its free port) onto the free destination port. (b) TELEGATE executes one remote CX through a cat-entangler: the target-side port acts as a proxy for the control q_c; neither qubit moves. (c) TELEGATEPACKET brackets an ordered run of remote CX gates sharing the control q_c between one cat-entangler/disentangler pair, amortising a single EPR pair over the whole run; each target is routed next to the port in turn while the ports stay reserved.

B. Communication Primitives

Three inter-core primitives are in play, sketched in Figure 3; each consumes exactly one EPR pair.

State teleportation (TELEDATA). Consumes one EPR pair to move a logical qubit's state from a physical position adjacent to a free source-side communication port to the (necessarily free) destination-side port in a neighbouring core [10]. The qubit then participates in local gates in its new core.

Gate teleportation (TELEGATE). Executes a single remote CX directly across two adjacent cores via a cat-entangler protocol [7], [8]: one EPR pair entangles the two communication ports, the remote CX is applied using the target-side port as a proxy for the control qubit, and the entanglement is consumed on disentangling. Neither logical qubit physically moves.

Same-control packets (TELEGATEPACKET). When several consecutive remote CX gates in the DAG front share the same control qubit and land in the same target core, PORTER amortises them onto a *single* cat-entangler resource: the entangling and disentangling steps bracket an ordered

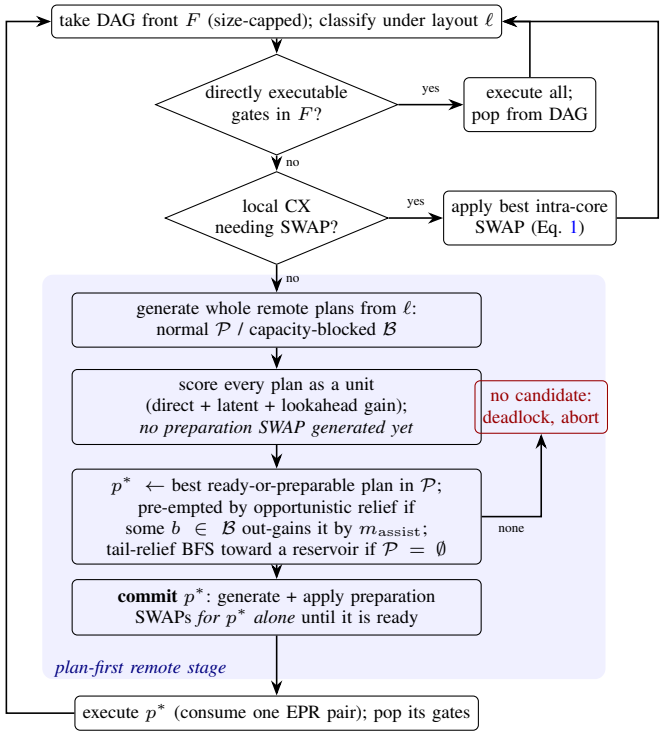


Fig. 4. One iteration of PORTER's compile loop (Algorithm 1). Local work always drains first; the shaded plan-first remote stage scores whole remote plans before any preparation SWAP exists, commits to one, and only then stages it.

sequence of remote CX and target-core 1Q gates, all executed while the two communication ports stay reserved. This is the same amortisation idea used by AutoComm [9], specialised to the shared-control case that arises naturally at the DAG front (Section II-A).

C. SABRE

SABRE [11] routes a single-core circuit by repeatedly selecting the SWAP that minimises a front-layer distance term plus a weighted extended-set lookahead term, both expressed as Δ (old distance minus new distance) on gates sharing a qubit with the candidate SWAP. PORTER inherits this scoring form for its intra-core stage (Section III-C) and extends the same direct/lookahead decomposition to its inter-core plan scorer (Section III-E).

III. The PORTER Router

Given a circuit and an initial layout ℓ_0 (which PORTER builds itself, Section III-A), PORTER repeats the following loop (Algorithm 1) until every gate has been emitted or a configurable iteration budget is exhausted. Figure 4 charts one iteration; its shaded region is the plan-first remote stage that distinguishes PORTER from flat-pool routers, and Section III-G walks through it on a concrete example.

A. Initial Layout

PORTER constructs its own starting layout ℓ_0 with a single deterministic, seed-free heuristic and does not require an external partitioner or a best-of-N search. It respects the capacity

Algorithm 1 PORTER `compile`(C, ℓ_0)

$\ell \leftarrow \ell_0$; initialise DAG active/frontier sets from C
while DAG has active gates **do**
  $F \leftarrow$ DAG front (bounded by `dag_front_limit`)
  classify F into singles, direct-local CX, local-CX-needing-SWAP, adjacent-remote CX, non-adjacent-remote CX *(Sec. III-B)*
  if singles or direct-local CX present **then**
    execute them all; remove from DAG; **continue**
  end if
  if local-CX-needing-SWAP present **then**
    generate candidate intra-core SWAPs, apply the best-scoring one (Eq. 1); **continue** *(Sec. III-C)*
  end if
  generate normal remote plans $\mathcal{P}$ and blocked-teleport plans $\mathcal{B}$ *(Sec. III-E)*
  if $\mathcal{P} \neq \emptyset$ **then**
    $p^* \leftarrow$ best ready-or-preparable plan in $\mathcal{P}$
    if some $b \in \mathcal{B}$ has gain $\gg$ gain(p^*) **then**
      $p^* \leftarrow$ best opportunistic capacity-relief plan for b *(Sec. III-H)*
    end if
  else
    BFS from each blocked target core to its nearest reservoir; $p^* \leftarrow$ best tail-relief plan *(Sec. III-H)*
  end if
  if p^* is `None` **then**
    report deadlock; abort
  end if
  apply p^*'s own preparation SWAPs until ready, then execute it
end while
return emitted operations, final layout, stats

invariant the routing loop depends on: every core keeps at least $f_{\min}$ free physical positions ($f_{\min}$=2), so that from the first iteration any core can serve as a TeleData destination and as a reservoir for capacity relief (Section III-H). A logical qubit is therefore never placed into a core that already holds $|c| - f_{\min}$ logicals, where $|c|$ is the core's physical capacity.

Temporal locality-greedy. The heuristic runs in three deterministic steps. *(i) Weighting.* Each CX gate $g = (q_i, q_j)$ contributes $\gamma^{\ell(g)}$ to the interaction weight $w(q_i, q_j)$ of its qubit pair, where $\ell(g)$ is g's as-soon-as-possible layer in the CX-only DAG and γ=0.99; early interactions thus weigh more than late ones, while repeated interactions on a pair accumulate. *(ii) Core assignment.* The logical qubits are visited in the order they first appear as a CX operand and each is placed in the core holding its heaviest already-placed neighbour when that core still has headroom above $f_{\min}$, falling back to the core with the most remaining reserved capacity otherwise. *(iii) Physical placement.* Within each core, the assigned logicals are laid onto physical positions by descending total interaction weight onto ascending intra-core centrality (sum of local distances), so the most-connected qubits take the most central slots; unused

slots are left idle.

This single construction subsumes what the earlier design split across two heuristics and a best-of-six search. The temporal discount front-loads the opening layer of interactions, recovering the first-CX-layer co-location a "balanced first-layer" scan gave; the accumulated multiplicity together with the first-appearance visit order tracks chain-structured circuits a first-layer scan cannot – a GHZ-style chain exposes only one disjoint pair in its first CX layer, yet successive chain neighbours share weight and are visited in sequence, so they land in the same core. Of the three steps the visit order carries the most weight: replacing the first-appearance order with a random permutation inflates geometric-mean EPR from 55.4 to 103.3 over the eighteen benchmark instances (Section IV), whereas replacing centrality placement with a random within-core shuffle costs only $55.4 \rightarrow 56.4$.

External layouts. When ℓ_0 instead comes from an outside mapper that may fill a core to capacity (Appendix B), PORTER first repairs it (`repair_full_cores`): it evicts one logical qubit from every saturated core into the core with the most spare capacity, restoring the $f_{\min}$ reservation before routing. A core packed to its physical capacity can never clear an occupied communication port with local SWAPs, so without this step its qubits would be unreachable by any teleport or telegate. PORTER's own heuristic reserves $f_{\min}$ by construction and never invokes the repair.

B. Front Classification

Every gate g in the (size-capped) DAG front is classified under the current layout ℓ: 1Q gates are always directly executable; a CX gate $g = (q_i, q_j)$ with $\kappa_\ell(q_i) = \kappa_\ell(q_j) = c$ (same core) is *direct-local* if $d_{\mathrm{L}}(c, \ell(q_i), \ell(q_j)) = 1$ and *local-needing-SWAP* otherwise; a CX gate with $\kappa_\ell(q_i) \neq \kappa_\ell(q_j)$ is *adjacent-remote* if the two cores are core-graph neighbours and *non-adjacent-remote* otherwise. Directly executable gates are always drained first and completely, so remote-stage decisions are only made once the front contains solely gates that need routing.

C. Intra-Core SWAP Scoring

For a local-needing-SWAP gate set F_{loc}, PORTER builds candidate SWAP edges from the coupling-graph neighbours of every physical position touched by F_{loc}, simulates each SWAP, and scores it as

$$J(a) = \bar{d}_{\mathcal{F}}(a) + \lambda_{\mathrm{loc}}\, \bar{d}_{\mathcal{E}}(a), \qquad \lambda_{\mathrm{loc}} = 0.8, \tag{1}$$

the mean post-SWAP local distance over F_{loc} plus a weighted mean over an extended set $\mathcal{E}_{\mathrm{loc}}$: up to N_{loc}=20 local CX gates collected from the next L_{loc}=12 DAG layers via a virtual frontier simulation (Section IV-A). Ties are broken first by the raw front-layer distance, then, if still tied, by a decision uniformly sampled from a seeded random-number generator, with every seeded tie-break recorded in `RouterStats` for reproducibility auditing.

D. Plan-First Remote Stage

PORTER manipulates remote work as *plans*, and the word is chosen deliberately. A plan is a complete, committable specification of one inter-core action: which primitive fires (TeleData, TeleGate, or TeleGatePacket), which gate or same-control gate run it resolves, its direction (which endpoint moves, or which side hosts the control), and consequently its *readiness conditions* — which logical endpoint must sit adjacent to which communication port, and which ports must be free. What a plan deliberately leaves unspecified is the SWAP sequence that will establish those conditions. This is the classical deliberation-versus-execution split: alternatives are compared at the level of *what* remote work to do next, each scored as if its preconditions were met, and only the one committed plan is handed to the preparation machinery that works out *how*. A flat action pool, by contrast, makes the same choice implicitly and one micro-move at a time: every staging SWAP it selects is also a vote for whichever remote action that SWAP happens to serve, votes for competing actions interleave, and staging effort is routinely spent on actions that are never taken (Section III-G gives a concrete front where this changes the outcome).

When the front contains only remote CX gates, PORTER generates two disjoint candidate sets from the current layout:

- **Normal plans** $\mathcal{P}$: for every adjacent-remote CX gate, a single TeleGate; if a same-control run of ≥ 2 remote CX gates can be collected by simulating the DAG front forward (Section III-F), a TeleGatePacket; and, for either endpoint, a TeleData plan into the partner's core *if* that core already has at least `min_free_before_teleport` ($f_{\min}$=2) free physical positions. For non-adjacent-remote CX gates, only distance-reducing TeleData hops (one core-graph step closer to the gate's other endpoint) with sufficient destination capacity are generated.
- **Blocked plans** $\mathcal{B}$: the same directional TeleData candidates, but for destination cores that currently fail the $f_{\min}$ capacity test. These are never selected directly; they exist to detect when *no* normal plan exists because every candidate destination is saturated, or when a normal plan is technically available but a severely blocked plan promises much higher gain if capacity is relieved first.

Each plan is scored (Section III-E) *before* any preparation SWAP for it is generated. The router then:

1) selects the best plan in $\mathcal{P}$ that is either already physically ready or admits at least one improving preparation SWAP right now (see the preparation rule below);
2) compares its gain against the best plan in $\mathcal{B}$; if the blocked plan's gain exceeds the selected normal plan's by more than a fixed margin m_{assist}=5.0, the router instead searches for an *opportunistic capacity-relief* plan for the blocked plan's saturated destination core (Section III-H) and, if one exists, executes that instead;
3) if $\mathcal{P} = \emptyset$ (every remote CX gate is non-adjacent), the router runs the same breadth-first capacity-relief search directly on $\mathcal{B}$'s destination cores, grouped by hop count, taking the shortest-hop group with a viable candidate.

Once a plan is committed, the router repeatedly generates and applies *only that plan's* preparation SWAPs until it becomes ready, no further improving preparation SWAP exists, or the full decision state (plan identity, layout, RNG state) repeats — the latter raising a hard error rather than looping forever, bounded by `max_prep_iterations_per_plan` (10^4).

Preparation SWAPs. A committed plan's readiness requirements are exactly two *(core, neighbour-core, endpoint)* triples (source and destination side for TELEDATA; control and target side for TELEGATE/TELEGATEPACKET, whose target-side routing is deferred to execution time, Section III-F). For each unmet requirement, candidate preparation SWAPs are restricted to edges incident to either the plan's logical endpoint or a nearest free physical position tied for minimum distance to the relevant communication port (only while that port remains occupied); each candidate is scored by the post-SWAP value of the same readiness-cost function used to rank blocked plans (Section III-E), and at most one improving SWAP per core side is applied per pass.

E. Plan Scoring: Direct, Latent, and Lookahead Gain

Every plan's score is $\mathrm{total} = \mathrm{direct} + \mathrm{lookahead}$, computed differently per plan kind but sharing one Δ-form convention: for a CX gate $g = (q_i, q_j)$ and a candidate qubit relocation, $\Delta(g) = d_C(\kappa(q_i), \kappa(q_j)) - d_C(\kappa'(q_i), \kappa'(q_j))$, the reduction in core-graph distance between g's two cores.

TELEGATE contributes a constant direct gain of 1 (one remote CX resolved) and no lookahead term.

TELEGATEPACKET contributes a direct gain equal to the number of remote CX gates in the collected packet – every gate in an accepted packet is current, not speculative, work, since the whole packet executes under one cat-entangler resource once started.

TELEDATA decomposes its influence into three tiers, all computed by simulating the DAG frontier under the hypothetical post-teleport core assignment (Section IV-A):

- *Direct*: front-layer CX gates already involving the moving qubit, plus the seed gate itself.
- *Latent-direct*: CX gates on the moving qubit that appear in the simulated DAG front while the simulator can still make virtual progress with only 1Q gates and now-local CX gates (i.e. gates whose post-teleport cores coincide); these gates are current pressure, not prediction, because the candidate teleport directly determines their post-move core distance the moment they surface.
- *Lookahead*: CX gates on the moving qubit found in up to L_{tp}=16 further simulated DAG layers (capped at N_{tp}=50 gates) beyond the point where the latent-direct simulation naturally stops (no further gate is virtually executable). Layer k's contribution is weighted by $\max(w_{\min}, w_0\gamma^{k-1})$ with w_0=0.9, γ=0.9, $w_{\min}$=0.5.

$$\mathrm{direct} = \sum_{g\in\mathrm{direct}} \Delta(g) + w_{\mathrm{ld}} \sum_{g\in\mathrm{latent}} \Delta(g), \tag{2}$$

$$\mathrm{lookahead} = \sum_{(g,k)\in\mathrm{la}} w_k\,\Delta(g), \tag{3}$$

TABLE I
REMOTE PLANS GENERATED FOR THE FRONT OF FIGURE 5, WITH TOTAL GAINS (SECTION III-E).

Plan	Set	Gain	Readiness	Outcome
TELEGATEPACKET(q_0; $g_1 g_2 g_3$)	$\mathcal{P}$	3.0	1 prep SWAP	**committed**
TELEGATE(g_4)	$\mathcal{P}$	1.0	ready now	dominated
TELEGATE(g_1)	$\mathcal{P}$	1.0	1 prep SWAP	dominated
TELEDATA($q_1 \to c_0$)	$\mathcal{P}$	1.0	prep needed	dominated
TELEDATA($q_5 \to c_0$)	$\mathcal{P}$	1.0	prep needed	committed later
TELEDATA($q_0 \to c_1$)	$\mathcal{B}$	$\approx$2.1	blocked	margin $< m_{\mathrm{assist}}$
TELEDATA($q_4 \to c_1$)	$\mathcal{B}$	1.0	blocked	dominated

with w_{ld}=1.0 (`teleport_latent_direct_weight`). A capacity-risk term $1/\max(1, f_{\mathrm{dst}}-1)$ (post-move free count in the destination core) is tracked alongside the score for use as a tie-break dimension but does not enter total directly; plans are primarily ranked by total gain (descending), then readiness, then preparation cost, then plan kind (TELEDATA $\succ$ TELEGATEPACKET $\succ$ TELEGATE), then capacity risk, then lookahead gain, with a fully deterministic stable key as the final tie-break before any seeded random choice.

F. Same-Control Packet Collection

Starting from a seed adjacent-remote CX gate $g = (q_c, q_t)$ with control core c_c and target core c_t, PORTER simulates the DAG frontier forward for up to `extended_packet_window` (120) layers, greedily absorbing into the packet: (i) any front CX gate sharing control q_c whose target is also in c_t, up to `max_packet_size` (8) such gates; and (ii) any front 1Q gate whose qubit is already in c_t, since these execute freely while the packet's cat resource is live. Collection stops at the first layer that contributes nothing, or once the size cap is reached. Because this DAG-frontier simulation admits gates from later generations that are not literally in the (bounded) current DAG front, a packet may contain gates unreachable by a naively front-limited scan; the collected order is preserved and replayed exactly at commit time, as one `cat_entangler` (consuming the packet's single EPR pair), one `cat_cx` per member — each target routed to its communication port by ordinary local SWAPs, forbidden from touching the two live ports — and a closing `cat_disentangler`.

G. Running Example: Why Commit to a Plan?

Figure 5 shows a front where plan commitment, the packet vocabulary, and the $f_{\min}$ capacity gate all interact, and where PORTER's decision differs from a flat action pool's. Two adjacent cores of architecture A are shown. Core c_0 holds q_0, q_4 (adjacent to the port), and q_6, with six free positions; core c_1 holds eight qubits — the targets q_1, q_2, q_3, the port-adjacent q_5, and four spectators — so its *only* free position is its port. The remaining circuit fragment is g_1=CX(q_0, q_1), g_2=CX(q_0, q_2), g_3=CX(q_0, q_3), g_5=CX(q_0, q_6), chained on q_0, plus an independent g_4=CX(q_4, q_5). The front is therefore $\{g_1, g_4\}$ — g_2 and g_3 share q_0 with g_1 and only surface after it executes. All front gates are adjacent-remote, so the plan-first stage fires.

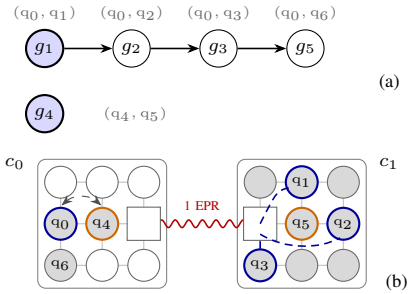


Fig. 5. Running example. (a) Remaining circuit fragment as a DAG; the front (shaded) is $\{g_1, g_4\}$. (b) Layout on two adjacent cores of architecture A. Blue circles are the committed TELEGATEPACKET's participants (q_0 and the targets q_1, q_2, q_3, reached by cat-CXs, dashed); orange circles are the endpoints of the competing, already-ready single TELEGATE for g_4; grey circles are spectators. Core c_1's only free position is its port, so $\text{free}(c_1) = 1 < f_{\min}$ and every TELEDATA into c_1 is capacity-blocked. The dashed double arrow is the packet's one preparation SWAP, which moves q_0 next to the port.

What PORTER does. Table I lists the generated plans. Seeding packet collection at g_1, the frontier simulation absorbs g_2 and g_3 (same control q_0, targets in c_1): a three-gate packet with direct gain 3. Both single telegates and all TELEDATA plans into c_0 gain ≈1. TELEDATA($q_0 \to c_1$) would gain more — direct 1 for g_1, latent-direct 2 for g_2, g_3, minus 0.9 of lookahead for g_5, whose partner q_6 stays in c_0 — but c_1 fails the $f_{\min}$ test, so it lands in $\mathcal{B}$; its gain ≈2.1 does not exceed the packet's 3.0 by m_{assist}=5, so no capacity relief pre-empts. Plans are ranked by gain first — readiness is only a tie-break — so the *unready* packet (gain 3) beats the *ready* telegate for g_4 (gain 1). PORTER commits the packet, generates its one preparation SWAP (q_0 next to the port), then emits one cat-entangler, three cat-CXs (each target routed beside the port in turn), and the disentangler: *one* EPR pair for g_1, g_2, g_3. The packet's completion exposes g_4 and g_5 together; the locality-first stage drains the now-local g_5 first (one SWAP plus a local CX), and the remote stage then re-generates plans for g_4: the single telegate and TELEDATA($q_5 \to c_0$) now tie at gain 1, and the plan-kind tie-break (TELEDATA ≻ TELEGATE, Section III-E) commits the teleport — q_5 lands on c_0's port and g_4 finishes as a local CX. Two EPR pairs for the whole fragment. This is not a stylised trace: it is exactly the operation stream the implementation emits on this instance, replayed and certified by the independent verifier of Appendix D (full trace in Figure 6).

What a flat pool does. A router that scores individual actions — SWAPs and inter-core operations in one pool — fires the zero-cost ready telegate for g_4 first. The deeper problem is the staging SWAP the packet needs: exchanging q_0 and q_4 moves g_1's control one step closer to the port while moving g_4's one step away, so under a front-averaged distance metric of the kind used by SABRE descendants its score is ≈0; the information that makes this SWAP worth applying — three remote CXs behind g_1 amortised onto one EPR pair — is a property of a *whole plan*, invisible at single-action granularity. Vocabulary compounds the gap: g_2 and g_3 enter the front one at a time, so TELESABRE pays one telegate each, four EPR pairs for the fragment; DSABRE, restricted to TELEDATA, cannot move q_0 into the saturated c_1 without first evicting a resident (its `_force_make_room`, one more teleport), and ferrying q_1, q_2, q_3 out to c_0 instead costs three EPR pairs before g_4 is even addressed. Commitment finally protects the choice once made: only the packet's preparation SWAPs are generated — no staging effort is spent on the six losing plans — and the committed plan cannot be pre-empted mid-preparation by a candidate that its own staging transiently improved, a churn a flat pool permits by construction.

H. Capacity Relief

Tail-relief BFS. When every remote plan is capacity-blocked, PORTER runs a breadth-first search from each blocked destination core to the nearest *reservoir* core – one with at least $f_{\min}$ free slots – optionally chained beyond one hop (`capacity_relief_chain_enabled`). Every logical qubit currently in the BFS path's penultimate core becomes a tail-relief TELEDATA candidate into the reservoir, scored and selected by the same plan machinery, grouped and tried in ascending hop-count order.

Opportunistic relief. Even when a normal plan is available, a severely blocked plan can still win if relieving its target core's capacity would unlock substantially more gain (Section III-D); this differs from tail-relief BFS in firing pre-emptively, before any plan is actually starved, whenever the margin m_{assist} is exceeded.

I. Computational Complexity

Let N be the circuit's gate count, n its logical-qubit count, K the number of cores, and M the per-core physical-qubit count ($P{=}KM$ total physical qubits). Hardware distance tables (d_{L} per-core, d_{C} core-graph) are precomputed once at construction, in $O(KM^2 \log M + K^2 \log K)$, matching the standard two-level Dijkstra decomposition used by comparable routers [5]. Per iteration, front classification and direct-execution draining cost $O(|F|)$; intra-core SWAP scoring evaluates $O(M)$ candidate edges each at $O(|F_{\text{loc}}| + N_{\text{loc}})$ via one $O(|\text{frontier}|)$-cost DAG snapshot, reused across candidates; remote-plan scoring evaluates $O(n)$ candidates each at $O(N_{\text{tp}})$ via the same amortised snapshot cost. Free-slot-queue maintenance is $O(1)$ amortised per layout mutation since only the touched core(s) are refreshed (Section IV-A). Summing over N iterations gives $T_{\text{total}} = O\big(N(Mn + KM^2 \log M)\big)$ excluding the one-time precomputation term, in line with the per-iteration cost of other core-local distributed routers [5], [6].

IV. EXPERIMENTAL EVALUATION

A. Implementation and Verification

PORTER is implemented in Python and is publicly available, together with the verifier and comparison harness used below (Section I). The implementation is fully deterministic: for a fixed (circuit, hardware, random seed) triple the emitted operation stream is byte-identical across runs, and every seeded tie-break is counted in the emitted statistics, so behaviour preservation across refactorings can be certified

by hashing the serialized operation stream. Compile time is kept low by incremental data structures — chiefly the `DagFrontierSimulator` shared by extended-set collection and teleport lookahead (Sections III-C–III-F and III-E), a copy-on-write view over the live DAG state that tracks a private overlay of predecessor-count decrements against the shared `active`/`pred_base` maps, so that a snapshot costs $O(|\text{frontier}|)$ rather than $O(|\text{remaining circuit}|)$ and several independent virtual simulations can run from the same live DAG state without interfering — together with incremental DAG-frontier maintenance, cached hardware distance tables, $O(1)$ free-slot accounting, and apply/restore SWAP scoring in place of layout cloning. These reduce wall-clock compile time by roughly $4\times$ relative to a straightforward implementation of the same algorithm, with byte-identical output; Section IV-G reports absolute compile times.

Every reported run is certified by an independent replay verifier that treats the router as a black box: from the emitted operation stream, the original circuit, and the initial layout alone it re-derives gate coverage (each gate exactly once), per-qubit dependency order, coupling-edge legality, the port and isolation rules of all three communication primitives, and consistency of the emitted statistics. The verifier passes on 60 randomized circuits and on every instance in the evaluation below, and is itself validated by fault injection; Appendix D details its checks, a certified operation-by-operation trace of the running example, and the fault-injection study.

B. Setup

Architectures. We use the same two grid-of-grids topologies introduced by TELESABRE [6] and re-used by DSABRE [5]: architecture A, a 2×2 array of 3×3 cores (36 physical qubits, 4 inter-core links), and architecture H, a 2×3 array of 4×4 cores (96 physical qubits, 7 inter-core links). Both are instantiated identically for all three routers.

Benchmarks. We use the same MQT-Bench [12] circuit files DSABRE's own evaluation uses (confirmed by exact CX-count match against its published tables): a 25-qubit suite (AE, GHZ, Graphstate, QFT, QNN, Random) on architecture A, and a 64-qubit suite (the same six families) on architecture H, with QNN using the deeper 8,126-CX variant. To test at intermediate scale on the same architecture H we add an 80-qubit suite of the same six families, generated with `mqt.bench` 2.2.2. The 80-qubit AE and QNN instances cannot be produced by the stock generator and are reconstructed to match its own construction exactly; in particular, QNN is the deep 12,718-CX ZZ-feature-map circuit rather than the degenerate 79-CX chain the current generator emits (whose interaction graph is identical to GHZ's). Appendix A details both reconstructions and their byte-level validation. Table II lists CX counts.

Baselines. We compare against TELESABRE [6] and DSABRE [5]. For DSABRE we report only `dSABRE_BurstExt`, its BFS-layer extended-set variant and the project's own default/production router (its topological-extended-set ablation, DSABRE-Topo, is omitted). TELESABRE's stock reference C binary crashes deterministically on architecture A, whose 9-slot cores are materially smaller than the 4×4 cores it was originally evaluated on: its Hungarian initial-layout pass can leave a logical qubit unassigned when no core's remaining capacity exceeds its free-slot threshold, and the unassigned sentinel is then used as a negative array index. All TELESABRE numbers in this paper use a minimally patched binary that falls back to the least-loaded core in that case; the patch and a detailed account of the issue are provided in our public repository.

TABLE II
BENCHMARK CIRCUIT CX COUNTS (MEASUREMENTS/BARRIERS STRIPPED).

Circuit	25q (arch. A)	64q (arch. H)	80q (arch. H)
AE	558	1962	2780
GHZ	24	63	79
Graphstate	25	64	80
QFT	580	1966	2660
QNN	1223	8126	12718
Random	1124	1627	23381

Compiler timeout. Every individual compiler invocation – one TELESABRE subprocess run, one DSABRE routing-pass call, or one PORTER `compile` call – is capped at 900 s wall-clock; a call that exceeds it is recorded the same way as any other failure to complete (missing entry, excluded from that protocol's geometric mean). This is a harness-level bound applied uniformly across all three routers and all three protocols below, not specific to any one circuit.

C. Initial-Layout Protocols

We evaluate under three initial-layout protocols, designed to separate routing quality from initial-layout quality.

Own-protocol comparison. Each router is first evaluated under its own established best-practice initial-layout protocol. For the two baselines this is best-of-three, matching the convention already used to compare TELESABRE and DSABRE to each other [5]; for PORTER it is a single deterministic run:

- **TELESABRE**: the compiled C binary's built-in hungarian initial-layout heuristic, config seeds 0–2, best EPR count of the successful runs.
- **DSABRE**: Qiskit `SabreLayout` on the corners-removed architecture graph (seeds 0–2, with the two least useful corners of *every* core removed — see below) followed by SABRE-style forward → backward → forward routing passes, best EPR count.
- **PORTER**: a *single* routing pass from its single deterministic temporal locality-greedy layout (Section III-A), router seed 0. PORTER performs no layout search and no seed sweep, so its column reports one attempt per circuit where each baseline's reports the best of three; every number is exactly reproducible from the circuit file alone.

DSABRE's corner removal reserves low-degree physical qubits as communication slots by excluding them from the SabreLayout coupling map. Its published selection takes the four lowest-degree vertices of the *whole-chip* graph; on both of our architectures, whose physical-qubit ids are assigned

TABLE III
HARDWARE COST AND HEURISTIC PARAMETERS (SEE SECTION III).

Parameter	Symbol	Value
Min. free before TELEDATA	$f_{\min}$	2
Opportunistic-relief margin	m_{assist}	5.0
Local-SWAP extended-set weight	λ_{loc}	0.8
Local extended-set max layers	L_{loc}	12
Local extended-set max CX	N_{loc}	20
Packet collection window (layers)	–	120
Max packet size (remote CX)	–	8
TeleData lookahead max layers	L_{tp}	16
TeleData lookahead max CX	N_{tp}	50
Lookahead weight / decay / floor	$w_0, \gamma, w_{\min}$	0.9, 0.9, 0.5
Latent-direct weight	w_{ld}	1.0

core-by-core, these four vertices are, by construction, the four corners of a single core (core 0 in our numbering), so the remaining cores receive no reserved slots at all and regularly start fully packed. We strengthen the baseline's protocol here: we instead remove, in *every* core, the two corners farthest (by total shortest-path distance in the chip graph) from all other physical qubits — the least useful positions, whose reservation costs the mapper the least. Two per core is also the natural budget: removing all four corners of every core would leave architecture A only $4 \times (9 - 4) = 20$ usable slots, too few for its 25-qubit circuits, whereas two leave $4 \times (9 - 2) = 28$ there and $6 \times (16 - 2) = 84$ on architecture H, enough for the 80-qubit suite. We still do not reuse these SabreLayout candidates for PORTER in this comparison, to keep each router's own-protocol result tied to its own established initial-layout heuristic.

Shared-layout comparison. To isolate routing quality from initial-layout quality, we additionally route both DSABRE and PORTER from TELESABRE's *own* initial layout: for the winning seed of the run above, we extract, from TELESABRE's optional JSON trace, the physical-to-logical mapping recorded at iteration 0 – i.e. the layout TELESABRE itself settled on after its internal forward/backward/forward layout refinement, immediately before its accepted forward routing pass – and feed that identical mapping, unmodified, to a single DSABRE run and a single PORTER run (no per-router layout search; the layout is now fixed and given). Unlike DSABRE's own SabreLayout, TELESABRE's hungarian heuristic reserves free-slot headroom per core by construction (tuned to its own teleport threshold), so it is compatible with PORTER's teleport precondition without adjustment on every circuit we tested.

A third, independent layout source — a placement generated by `pytket-dqc`'s `HypergraphPartitioning` allocator [13], routed by both DSABRE and PORTER — is reported and discussed in Appendix B.

D. Own-Protocol Results

TELESABRE's EPR count is teledata plus telegate operations; DSABRE's is teledata only; PORTER's is teleport plus telegate plus telegate-packet operations (one EPR pair per packet, regardless of how many remote CX gates it amortises).

TABLE IV
25-QUBIT SUITE, OWN-PROTOCOL COMPARISON (ARCHITECTURE A, 2×2 CORES OF 3×3, 36 PHYSICAL QUBITS). TELESABRE AND DSABRE ARE BEST-OF-3 SEEDS; PORTER IS A *single* RUN FROM ITS SINGLE DETERMINISTIC LAYOUT (SECTION IV-C). Δ_{TS} IS PORTER'S EPR CHANGE RELATIVE TO TELESABRE; NEGATIVE IS BETTER.

	TELESABRE		DSABRE		PORTER		
Circuit	EPR	SWAP	EPR	SWAP	EPR	SWAP	Δ_{TS}
ae	118	370	94	399	65	328	−44.9
ghz	11	30	4	18	4	29	−63.6
graphstate	17	43	4	19	7	34	−58.8
qft	141	371	91	462	74	327	−47.5
qnn	357	770	155	790	119	642	−66.7
random	632	1512	431	1521	401	1327	−36.6
gmean (6)	94.3		45.7		**43.1**		**−54.3**

TABLE V
64-QUBIT SUITE, OWN-PROTOCOL COMPARISON (ARCHITECTURE H, 2×3 CORES OF 4×4, 96 PHYSICAL QUBITS). METHODOLOGY AND NOTATION AS TABLE IV. *TELESABRE FAILS TO CONVERGE WITHIN 3 SEEDS; GMEAN IS OVER THE 5 CIRCUITS ON WHICH IT DOES.

	TELESABRE		DSABRE		PORTER		
Circuit	EPR	SWAP	EPR	SWAP	EPR	SWAP	Δ_{TS}
ae	323	1508	244	1863	177	1627	−45.2
ghz	11	111	14	73	6	86	−45.5
graphstate	76	379	18	113	20	131	−73.7
qft	410	1872	240	2252	262	1846	−36.1
qnn	1365	6292	539	8394	864	6595	−36.7
random*	—	—	738	3827	737	4048	—
gmean (5)	172.1		95.5		**86.4**		**−49.8**

All three report in the same one-EPR-pair unit. SWAP counts intra-core SWAPs only.

Across the three suites the own-protocol comparison supports one headline claim: *a single* PORTER *run from its single deterministic layout posts the lowest geometric-mean EPR count of the three routers on every suite*, despite each baseline reporting the best of three attempts – 43.1 vs. DSABRE's 45.7 and TELESABRE's 94.3 at 25 qubits, 86.4 vs. 95.5 and 172.1 at 64, and 14.0 vs. 21.0 and 40.9 at 80 (the 80-qubit gmeans are over the two circuits TELESABRE completes; the fuller circuit-by-circuit picture, including two circuits on which DSABRE wins, is below). It is also the only router to complete all eighteen instances (DSABRE completes seventeen, TELESABRE thirteen). The per-suite detail follows.

25-qubit suite. PORTER reduces geometric-mean EPR consumption by **54.3%** against TELESABRE, with per-circuit reductions from −36.6% (Random) to −66.7% (QNN) and no regressions. Against DSABRE it wins AE, QFT, QNN, and Random outright, ties GHZ (both 4), and concedes only Graphstate (7 vs. 4) – the extreme sparse-partitionable case where SabreLayout on the corners-removed architecture graph, refined over three forward–backward–forward routing passes, finds a near-ideal core partition before routing begins, exactly the regime DSABRE's own evaluation identifies as its strongest [5]. On the CX-dense circuits the margin runs the

TABLE VI
80-QUBIT SUITE, OWN-PROTOCOL COMPARISON (ARCHITECTURE H, SAME PHYSICAL TOPOLOGY AS THE 64-QUBIT SUITE, MORE LOGICAL QUBITS PER CIRCUIT). METHODOLOGY AND NOTATION AS TABLE IV. *TELESABRE FAILS TO CONVERGE; GMEAN IS OVER THE 2 CIRCUITS ON WHICH IT DOES. QNN HERE IS THE RECONSTRUCTED DEEP ZZ-FEATURE-MAP QNN (12,718 CX, APPENDIX A), NOT THE DEGENERATE STOCK CHAIN.

	TELESABRE		DSABRE		PORTER		
Circuit	EPR	SWAP	EPR	SWAP	EPR	SWAP	Δ_{TS}
ae*	—	—	506	3438	490	3156	—
ghz	19	172	20	143	7	115	−63.2
graphstate	88	521	22	131	28	198	−68.2
qft*	—	—	670	3844	446	2968	—
qnn*	—	—	811	13960	1517	10920	—
random*	—	—	—	—	11040	62119	—
gmean (2)	40.9		21.0		**14.0**		**−65.8**

TABLE VII
SHARED-LAYOUT COMPARISON, ALL SUITES: DSABRE AND PORTER ARE BOTH ROUTED, SINGLE-SHOT, FROM TELESABRE'S OWN INITIAL LAYOUT (SECTION IV-C). Δ_{TS} COLUMNS GIVE EACH ROUTER'S EPR CHANGE RELATIVE TO TELESABRE'S OWN-PROTOCOL RESULT; NEGATIVE IS BETTER. ROWS EXIST ONLY WHERE TELESABRE CONVERGES AND THUS PROVIDES A LAYOUT TO SHARE: ALL SIX CIRCUITS AT 25Q, FIVE OF SIX AT 64Q (RANDOM MISSING), TWO OF SIX AT 80Q (GHZ AND GRAPHSTATE); PER-SUITE GMEANS ARE OVER THE AVAILABLE ROWS.

		TELESABRE		DSABRE			PORTER		
Suite	Circuit	EPR	SWAP	EPR	SWAP	Δ_{TS}	EPR	SWAP	Δ_{TS}
25q	ae	118	370	77	393	−34.7	62	267	−47.5
25q	ghz	11	30	8	33	−27.3	6	26	−45.5
25q	graphstate	17	43	10	37	−41.2	14	57	−17.6
25q	qft	141	371	89	410	−36.9	96	343	−31.9
25q	qnn	357	770	177	818	−50.4	158	707	−55.7
25q	random	632	1512	439	1529	−30.5	412	1392	−34.8
	gmean (6)	94.3		59.1		**−37.3**	56.5		**−40.1**
64q	ae	323	1508	183	1963	−43.3	233	1753	−27.9
64q	ghz	11	111	8	124	−27.3	8	104	−27.3
64q	graphstate	76	379	32	167	−57.9	49	289	−35.5
64q	qft	410	1872	228	2104	−44.4	227	1668	−44.6
64q	qnn	1365	6292	544	8250	−60.1	679	5963	−50.3
	gmean (5)	172.1		89.7		**−47.9**	107.1		**−37.8**
80q	ghz	19	172	15	173	−21.1	11	156	−42.1
80q	graphstate	88	521	55	306	−37.5	71	402	−19.3
	gmean (2)	40.9		28.7		**−29.8**	27.9		**−31.7**

other way (QNN: 119 vs. 155; AE: 65 vs. 94), consistent with PORTER's TELEGATEPACKET primitive amortising runs of shared-control remote CX gates that a teledata-only router must pay for individually (Section IV-F isolates this effect).

64-qubit suite. TELESABRE fails to converge within three seeds only on Random here; on QNN (8,126 CX) it succeeds at 1,365 EPR pairs and 6,292 SWAPs, exactly matching DSABRE's own published number for this circuit [5] – a useful cross-check that our harness reproduces TELESABRE faithfully. The gmean row is computed over the five circuits on which TELESABRE converges. PORTER improves on TELESABRE on every converged circuit (−36.1% to −73.7%); notably, the chain-structured GHZ that defeats locally-greedy schedulers generally – DSABRE regresses against TELESABRE on it both in our reproduction (14 vs. 11) and in its own published table [5] – is now a PORTER *win* (6 EPR pairs vs. 11 and 14): the first-appearance visit order of the temporal locality-greedy layout places successive chain neighbours in the same core, dissolving the pathology at the layout stage rather than fighting it at routing time. Against DSABRE, PORTER takes the gmean (86.4 vs. 95.5), winning AE (177 vs. 244), GHZ, and Random (737 vs. 738, where TELESABRE completes neither); DSABRE retains Graphstate (18 vs. 20), QFT (240 vs. 262), and, most clearly, the deep QNN (539 vs. 864). That QNN row is PORTER's weakest result in any suite and an honest limit of the single-layout policy: it is the one benchmark where the deterministic layout is decisively worse than alternatives PORTER itself can produce (its retired best-of-six search reached 642 there), suggesting layout sensitivity that a future refinement pass could recover.

80-qubit suite. TELESABRE converges on only two of the six circuits here (GHZ and Graphstate; Table VI), so its gmean basis is thin. PORTER wins that basis (14.0 vs. DSABRE's 21.0, TeleSABRE's 40.9), driven by GHZ (7 EPR pairs vs. 20 and 19). Completion is the clearer story at this scale: PORTER finishes all six circuits, DSABRE five (missing Random), TELESABRE two. Over the five circuits DSABRE also completes, PORTER wins three (AE 490 vs. 506, GHZ 7 vs. 20, QFT 446 vs. 670) and loses two (Graphstate 28 vs. 22 and, decisively, the deep QNN 1,517 vs. 811), for a geometric mean of 145.4 against DSABRE's 164.6.

The deep QNN deserves emphasis because it corrects a defect in the original suite: at 80 qubits the stock `mqt.bench` QNN is a 79-CX chain *identical* to GHZ (Appendix A), which would have double-counted the easy chain case and hidden the hard one. The reconstructed 12,718-CX QNN is a fully-connected interaction graph, and on it PORTER loses to DSABRE (1,517 vs. 811) by the same mechanism as the 64q QNN (864 vs. 539): its single deterministic layout cannot match DSABRE's best-of-three SabreLayout partition on an all-to-all circuit. The two QNN rows are now a consistent, honest boundary of the single-layout policy – the regime a future layout- or route-refinement pass (Section VI) most needs to target.

On Random (23,381 CX, by far the largest circuit in any suite), PORTER is the only router to finish, in 37 s at 11,040 EPR pairs, while DSABRE exceeds the uniform 900 s per-call timeout (Section IV-B). Running DSABRE past that cap to characterize the failure shows it is not a deadlock: its `max_iterations=20000` budget (carried over from the 64-qubit suite per DSABRE's own scaling guidance [5]) is reached after 938 s of its first forward pass alone, having cleared 1,233 of the 23,381 CX gates with genuine forward progress – a resource-budget limitation on an unusually dense circuit (3–18× the CX/qubit ratio of any other benchmark), not an in-principle inability. Still, the practical cost of a budget large enough to finish (three layout candidates through up to three passes each) would run to hours against PORTER's 37 seconds.

E. Shared-Layout Results

The own-protocol comparison mixes two effects: the quality of each router's initial layout and the quality of its routing decisions. Table VII removes the first effect entirely – both DSABRE and PORTER route, single-shot, from the identical layout TELESABRE built for itself – so what remains is a pure routing-quality comparison on a mapping optimised for neither router.

25-qubit suite. Under TELESABRE's own initial layout, both DSABRE and PORTER still improve substantially on TELESABRE (-37.3% and -40.1% gmean respectively), with PORTER the better of the two on four of six circuits (AE, GHZ, QNN, Random) and on the gmean. DSABRE keeps Graphstate and QFT, the circuits its partition-then-refine strategy suits best under its own protocol too.

64-qubit suite. TELESABRE's initial layout is available for five of six circuits. DSABRE wins the gmean here (-47.9% vs. PORTER's -37.8%): it takes AE, Graphstate, and QNN outright, ties GHZ (both -27.3%), and is edged out only narrowly on QFT (228 vs. 227 EPR). On architecture H's larger, 16-slot cores, DSABRE's forward–backward–forward routing passes evidently extract more from a fixed externally supplied layout than PORTER's single forward pass does – the one setting in our evaluation where they do. PORTER nonetheless wins this suite end-to-end (Table V), because its own layout is enough better than the one being shared here to more than repay the difference; a backward-refinement pass over PORTER's route is the natural mechanism to close the residual gap (Section VI).

80-qubit suite. TELESABRE's initial layout can be extracted only for the two circuits on which it converges (GHZ and Graphstate), so this block is indicative rather than conclusive. PORTER edges DSABRE on the two-circuit gmean (-31.7% vs. -29.8%), winning GHZ (11 vs. 15 EPR pairs) while DSABRE keeps Graphstate. GHZ is a win for PORTER even from a layout it did not build, so its chain-circuit strength is a property of the router as well as of its layout.

Fixed-layout comparison, pooled. A second source of externally supplied layouts — `pytket-dqc`'s `HypergraphPartitioning` placements, routed by both remaining routers on the 25q and 64q suites — is reported in Appendix B. Pooling that grid with Table VII gives thirty instances on which DSABRE and PORTER routed the *identical* externally supplied layout: PORTER uses fewer EPR pairs on fifteen, DSABRE on twelve, with three ties, and PORTER's pooled geometric mean is lower (73.8 vs. 76.7). Whatever share of each router's end-to-end result its own layout contributes, the routing stages themselves are at worst at parity – and PORTER reaches that parity in a single forward pass, without DSABRE's forward–backward–forward schedule.

F. Mechanism Ablation

Table VIII attributes PORTER's EPR consumption to its two distinctive mechanisms by disabling each in isolation, holding the initial layout and every other parameter fixed (the six omitted rows – GHZ and Graphstate at every scale – form no packets and never trigger relief, so all three configurations coincide there).

TABLE VIII
MECHANISM ABLATION: PORTER EPR WITH THE TELEGATEPACKET PRIMITIVE DISABLED (`MAX_PACKET_SIZE=1`) AND WITH BOTH CAPACITY-RELIEF TIERS DISABLED, ALL FROM THE SAME TEMPORAL LOCALITY-GREEDY LAYOUT AT ROUTER SEED 0. "PK" IS THE PACKET COUNT OF THE FULL CONFIGURATION. †FAILS TO COMPLETE (DEADLOCK OR ITERATION-GUARD TRIP). THE GMEAN ROW IS OVER THE FOURTEEN INSTANCES ALL THREE CONFIGURATIONS COMPLETE.

Suite	Circuit	Full	No packets	No relief	pk
25q	ae	65	71	80	8
25q	qft	74	74	140	0
25q	qnn	119	122	212	0
25q	random	401	412	395	43
64q	ae	177	211	327	32
64q	qft	262	372	382	45
64q	qnn	864	931	600	240
64q	random	737	843	789	77
80q	ae	490	688	†	264
80q	qft	446	435	†	209
80q	qnn	1517	1571	†	448
80q	random	11040	11628	†	572
gmean (14)		57.6	61.4	67.0	

Packets. Disabling TELEGATEPACKET forces every shared-control remote CX run to pay per gate (single telegates or teleports). The cost tracks the packet count: $+42\%$ EPR on 64q QFT (262 $\rightarrow$ 372, 45 packets), $+40\%$ on 80q AE (490 $\rightarrow$ 688, 264 packets), $+19\%$ on 64q AE, $+14\%$ on 64q Random, and smaller gains on the packet-heavy deep-QNN rows (80q QNN 1,517 $\rightarrow$ 1,571, 448 packets), against a $+7\%$ overall gmean. The lone counter-move (80q QFT, $446 \rightarrow 435$) shows packet formation is a heuristic commitment that can occasionally displace a slightly better teleport plan, but the direction is overwhelmingly one-sided.

Capacity relief. Disabling both relief tiers (Section III-H) is not merely more expensive – it is *unsafe*: 80q AE, QFT, QNN, and Random all fail to complete, because with every candidate destination below $f_{\min}$ free slots and no mechanism to drain qubits toward reservoir cores, the deterministic-greedy loop has no legal remote plan left. Where routing does complete, dense circuits pay heavily ($+89\%$ on 25q QFT, $+78\%$ on 25q QNN, $+85\%$ on 64q AE). The one instance relief *hurts* (64q QNN, $864 \rightarrow 600$ without it) shows the opportunistic tier can overpay on a circuit whose congestion would resolve itself; tightening that comparator is a natural tuning opportunity. Overall, relief is what converts PORTER's raise-on-deadlock failure model (Section III-D) from a liability into a non-event: with both tiers active, no benchmark instance in any protocol deadlocks.

G. Compile Time

Table IX reports per-attempt wall-clock time: one `route` call for DSABRE (one SabreLayout candidate through its forward–backward–forward pass schedule), one `telesabre_run` attempt for TELESABRE, and one `compile` call for PORTER, all on the same machine. For

TABLE IX
PER-ATTEMPT WALL-CLOCK TIME (SECONDS) ON REPRESENTATIVE CIRCUITS; ONE CANDIDATE LAYOUT/SEED PER ROUTER. *DID NOT COMPLETE.

Circuit	CX	TELESABRE	DSABRE	PORTER
ae (25q)	558	0.16	1.33	0.07
random (25q)	1124	1.31	5.63	0.42
ae (64q)	1962	0.99	29.29	0.32
qft (64q)	1966	1.89	13.30	0.41
qnn (64q)	8126	21.54	129.94	2.23
random (80q)	23381	*	*	37.40

the baselines the own-protocol totals in Section IV-D are 3× these per-attempt figures (best-of-three); for PORTER the per-attempt figure *is* the total, since its protocol is a single run from a single deterministic layout. PORTER is consistently the fastest per attempt, and the gap widens with circuit size: on the 1,962-CX AE circuit it is 92× faster than DSABRE's single attempt (275× against the best-of-three total); on the 8,126-CX QNN circuit, 58×; on the 23,381-CX 80-qubit Random circuit, DSABRE's own-protocol attempt does not complete at all (Table VI) while PORTER finishes in 37 s. This reflects the performance engineering of Section IV-A as much as any algorithmic difference between the routers – DSABRE's reference implementation has not been subjected to the same optimisation pass – so it should be read as a statement about implementation maturity rather than an inherent complexity gap between the two approaches.

V. RELATED WORK

TELESABRE [6] is our closest architectural relative: both routers support teledata and gate teleportation and both build on the SABRE extended-set scoring idea. The two differ chiefly in decision structure: TELESABRE scores inter-core candidates via a contracted-communication-graph Dijkstra energy that mixes staging cost and lookahead into one flat action pool per candidate, whereas PORTER scores remote *plans* independently of their preparation cost and only generates preparation SWAPs for the one plan selected (Section III-D), and adds a same-control packet primitive that amortises runs of shared-control remote CX gates onto one EPR pair. PORTER's capacity handling is also more proactive: alongside TELESABRE's safety-valve iteration cap, PORTER maintains an explicit blocked-plan set and both a breadth-first tail-relief search and an opportunistic pre-emptive relief comparator (Section III-H).

DSABRE [5] (we compare against its default `dSABRE_BurstExt` variant, "DSABRE") is the stronger of the two baselines and the closest competitor on EPR count. Its strategy inverts PORTER's cost profile: a SabreLayout-based initial allocation refined through forward–backward–forward routing passes, run over three seeds, buys excellent core partitions – it remains the best router on the sparse, highly partitionable Graphstate family in every protocol we test – at a compile-time cost that is 58–92× PORTER's per attempt (Section IV-G). Under each router's own protocol PORTER's single run posts the lower geometric-mean EPR on all three suites (Section IV-D), and across thirty instances routed from identical externally supplied layouts, PORTER wins fifteen to DSABRE's twelve (Section IV-E); DSABRE's remaining advantage is concentrated in its multi-pass refinement of an externally supplied layout on large cores (Section IV-E) and on the dense all-to-all QNN circuits (64q and 80q), where its best-of-three partition beats PORTER's single deterministic layout. DSABRE restricts itself to teledata; PORTER follows TELESABRE in also supporting gate teleportation and extends it with same-control packets, which matters on circuits with long runs of shared-control remote CX gates (e.g. QFT-like phase-kickback patterns) where amortising many remote CX gates onto one EPR pair can outweigh a less-optimal core partition (Section IV-F).

Single-QPU qubit mapping. The front+extended-set scoring loop PORTER and both baselines build on originates with SABRE [11]; LIGHTSABRE [14] formalises the Δ-form SWAP score PORTER reuses for its intra-core stage (Eq. 1) and inter-core direct-gain term (Eq. 3).

Allocation and scheduling. Hypergraph-partitioning allocation methods [13], [15] and time-aware schedulers [16], [17] address the upstream allocation and downstream scheduling stages respectively. Both remain complementary to PORTER: its own temporal locality-greedy allocation is a one-pass greedy heuristic, not a global optimiser, and Appendix B shows a hypergraph partition can still improve the most partitionable circuits when its generation cost is acceptable.

VI. CONCLUSION

We presented PORTER, a plan-committed router for multi-core distributed quantum computers whose central design choice is to separate remote-plan *selection* from remote-plan *preparation*, in contrast to TELESABRE and DSABRE, which score inter-core candidates and their staging SWAPs together in one flat action pool: every legal teleport, telegate, and same-control packet candidate (including capacity-blocked variants) is scored on its physical and DAG-lookahead merits before any core-preparation SWAP is generated, and preparation effort is then spent exclusively on the one plan chosen. The router is paired with a two-tier capacity-relief mechanism (breadth-first tail relief when every plan is blocked, opportunistic pre-emptive relief when one is severely blocked) and a deterministic, seed-free temporal locality-greedy initial layout that co-locates qubits by temporally discounted interaction weight in first-use order.

The combination is deliberately minimal – one layout, one forward pass, no seed sweep, no rollback – and the evaluation shows the minimalism costs nothing on this benchmark family: a single PORTER run posts the lowest geometric-mean EPR count of the three routers on all three suites (54.3%, 49.8%, and 65.8% below TELESABRE; below DSABRE's best-of-three on each suite), completes all eighteen benchmarks where DSABRE completes seventeen and TELESABRE thirteen, and runs 58–92× faster per attempt than DSABRE. Across thirty instances routed from identical externally supplied layouts, PORTER's single forward pass matches or beats DSABRE's forward–backward–forward schedule (fifteen wins to twelve). The ablation attributes the gains concretely: packets

save up to 42% EPR on shared-control-heavy circuits, and capacity relief is what makes the deterministic-greedy design complete at all at 80 qubits. At the largest scale we tested (a 23,381-CX circuit), PORTER completed in 37 s where both baselines exceed a uniform 900 s per-call compiler timeout – for DSABRE a resource-budget limitation rather than a deadlock (Section IV-D), but a practical throughput gap of two orders of magnitude nonetheless. Every reported run was certified by an independent replay verifier that re-derives, from the emitted operation stream alone, gate coverage, dependency order, and physical legality.

Two limits are equally concrete. DSABRE remains the best router for extreme partitionable structure (Graphstate, in every protocol), and its best-of-three SabreLayout partition beats PORTER's single deterministic layout on dense all-to-all circuits: the deep 64q and 80q QNN rows (864 vs. 539 and 1,517 vs. 811 EPR pairs) are PORTER's weakest results and a consistent boundary of the single-layout policy. Natural next steps follow directly: a backward-refinement pass over PORTER's single forward route, a lightweight exchange-refinement of the temporal locality-greedy partition targeting the dense-QNN case, and extending the packet primitive beyond the shared-control case.

Artefact availability.: The PORTER implementation, the independent replay verifier, the cross-router comparison harness (including the patched TELESABRE binary and its documentation), and raw per-circuit JSON results are publicly available at https://github.com/ebony72/porter.

APPENDIX A
THE 80-QUBIT SUITE: RECONSTRUCTION DETAIL

The 80-qubit suite (Section IV-B) was generated with `mqt.bench` 2.2.2 (device-independent generation followed by transpilation to our basis, since the installed `mqt.bench`'s target-aware native-gate path raised an unrelated `qiskit` compatibility error on every circuit we tried). AE at this size is not directly producible by stock `mqt.bench`: its phase-estimation construction evaluates $\mathrm{Operator}(Q)^{2^j}$ for each evaluation bit j, and the accumulated floating-point non-unitarity trips `qiskit`'s unitarity check (`"Input matrix is not unitary"`) beyond roughly 27 evaluation qubits, which is why published AE files stop near 54 qubits. Since the Bernoulli operator Q is a single-qubit rotation, we instead supply the power analytically, $Q^{2^j} = R_Y(2^{j+1}\theta_p)$ (R_Y periodic in 4π), and otherwise run `mqt.bench`'s own native-gate transpilation unchanged. The CNOT sub-sequence—which alone determines routing—is decomposition-invariant in these rotation angles and is byte-identical to stock `mqt.bench` at every size the latter can build (verified at 10, 20, 25, and 28 qubits); only the routing-irrelevant single-qubit angles carry floating-point error at high bit indices.

QNN required an analogous reconstruction. Between `mqt.bench` 1.1.0 (which produced the deep 8,126-CX 64q QNN this suite uses) and the installed 2.2.2, the benchmark's definition was changed from a ZZ feature map (full all-pairs entanglement) plus a `RealAmplitudes` ansatz to a plain Z feature map (no entanglement) plus the same ansatz. The current generator therefore emits a near-linear 79-CX circuit at 80 qubits whose two-qubit interaction graph is *identical* to GHZ's — a degenerate duplicate, not a distinct benchmark. We restore the original deep QNN by rebuilding the 1.1.0 construction (ZZ feature map, `reps`=2, full entanglement; `RealAmplitudes`, `reps`=1) and transpiling it to our basis; at 64 qubits this reproduces the checked-in 1.1.0 file's CNOT sub-sequence byte-for-byte (identical ordered pair sequence, all $\binom{64}{2}$ interacting pairs, 8,126 CX), so the 80-qubit instance (a fully-connected $\binom{80}{2}$-edge interaction graph, 12,718 CX) is a faithful extrapolation of the same construction.

APPENDIX B
EXTERNAL LAYOUTS FROM PYTKET-DQC

As a third, independent layout source complementing the shared-layout protocol of Section IV-C, we route DSABRE and PORTER from a placement generated by `pytket-dqc`'s `HypergraphPartitioning` allocator [13] (best-of-3 KaHyPar seeds on a `NISQNetwork` matching our core graph), converted to a physical layout by the same topology-aware intra-core assignment DSABRE's own pytket benchmark uses [5]: each core's assigned logical qubits are placed by descending interaction-graph degree onto physical positions ordered by ascending intra-core centrality, then locally refined by a bounded pairwise-swap pass. TELESABRE is excluded for the same structural reason as in Section IV-C (no externally supplied layout support in its CLI). `HypergraphPartitioning` only decides which *core* each logical qubit goes to; the capacity it sees for a core is the length of that core's declared qubit list in the `NISQNetwork`. Following DSABRE's stock corner-removal convention (the four lowest-degree corner qubits of the whole-chip graph; Section IV-C), we withhold those four corner qubits from the cores' declared capacity and from the later physical-placement step within each core. This guarantees only a four-qubit margin *chip-wide*, not per-core, so a core can still end up fully packed; rather than skip the run in that case, PORTER's general safeguard against fully packed cores (Section III-A) legalizes the layout automatically before routing begins. DSABRE's teledata-only mechanism does not need a free port on the side it teleports *from* and was never affected by this precondition. Because `pytket-dqc` uses gate teleportation, not state teleportation, its own reported cost is in a different unit (e-bits per Steiner-tree-amortised hyperedge, not EPR pairs per qubit move) and is included only as context, following DSABRE's own established convention for this comparison [5].

Table X reports a complete grid: PORTER's full-core safeguard legalizes the three layouts that would otherwise violate its teleport precondition (25q QNN, 64q AE, 64q QNN), at the cost of the layout-quality hit an emergency eviction inevitably takes – most visibly on 64q QNN, where the evicted layout lands PORTER at 1,031 EPR pairs against DSABRE's 509, its worst relative result on this suite. Excluding that outlier, results split close to evenly: PORTER uses fewer EPR pairs on 25q Graphstate/QNN/Random and 64q AE/QFT, DSABRE on 25q AE/QFT and 64q Graphstate/Random, and

TABLE X
PYTKET-MAPPING SHARED-LAYOUT COMPARISON (25Q AND 64Q SUITES), CORNERS-REMOVED NISQNETWORK CAPACITY. PYTKET E-BITS IS HYPERGRAPHPARTITIONING'S OWN COST IN A DIFFERENT UNIT (CONTEXT ONLY, NOT EPR-COMPARABLE). PORTER'S GENERAL FULL-CORE SAFEGUARD LEAVES NO MISSING ENTRIES ON THIS SUITE.

Suite	Circuit	`pytket` e-bits	DSABRE		PORTER	
			EPR	SWAP	EPR	SWAP
25q	ae	78	67	339	73	348
25q	ghz	3	3	14	3	17
25q	graphstate	4	5	24	4	19
25q	qft	70	92	437	118	438
25q	qnn	148	168	848	127	703
25q	random	564	440	1482	395	1340
64q	ae	264	361	2243	223	1638
64q	ghz	5	5	74	5	62
64q	graphstate	10	8	68	10	60
64q	qft	307	320	2112	278	2021
64q	qnn	621	509	8215	1031	6538
64q	random	1020	750	3969	861	4384

the two tie on both GHZ rows; geometric-mean EPR across all twelve rows is within 1.4% of parity (DSABRE 69.2 vs. PORTER 70.2). Pooled with the TELESABRE-layout grid of Table VII, PORTER uses fewer EPR pairs on fifteen of the thirty fixed-layout instances, DSABRE on twelve, with three ties (Section IV-E).

The grid above feeds pytket-dqc's partition to PORTER as a controlled probe. The complementary pipeline question – offering the same partition to PORTER as an optional extra layout candidate beside its own, which lowers geometric-mean EPR by a further 16% on the most partitionable circuits but is kept opt-in on cost and robustness grounds – is a configuration study of PORTER itself rather than part of this comparison, and is deferred to Appendix C.

APPENDIX C
PYTKET-DQC AS AN OPT-IN PORTER LAYOUT CANDIDATE

Appendix B feeds pytket-dqc's partition to PORTER as a controlled probe; the same partition can also be offered to PORTER as one extra layout candidate beside its own (Section III-A). Since PORTER routes far faster than pytket-dqc partitions, this is nearly free on the routing side; taking the better of the two per instance lowers the eighteen-instance geometric-mean EPR from 116.8 to 98.6 (−16%), with the gains concentrated on the most partitionable circuits (Graphstate $7 \to 4$, $20 \to 10$, $28 \to 11$ across the three suites; 80q AE $490 \to 352$). We keep it *opt-in* rather than the default for three reasons. First, on the dense and Random circuits the external partition is often *worse* for PORTER (64q QNN $864 \to 1{,}031$, 64q Random $737 \to 861$), so the best-of simply retains PORTER's own layout there. Second, the partition itself is expensive – seconds to minutes, and on dense circuits more than PORTER's entire routing pass (309 s to partition 80q Random versus ≈ 37 s to route it; 16 s versus 0.6 s on 80q AE) – against a sub-millisecond cost for PORTER's own layout build, so adopting it by default would erase PORTER's compile-time advantage (Section IV-G). Third, it reintroduces the external-partitioner dependency PORTER's self-contained layout avoids, and unlike it can leave a core fully packed, relying on the full-core repair of Section III-A. PORTER therefore defaults to its own layout and treats an external partition as an optional refinement, selected only when EPR is at a premium and its generation time is acceptable.

APPENDIX D
INDEPENDENT REPLAY VERIFICATION

Beyond code review, we built an independent replay verifier that treats the router as a black box: it takes only the emitted `CompiledOp` stream, the original circuit, and the initial layout, and re-derives, from scratch, whether the stream is a faithful and physically legal execution. The verifier maintains its own physical-position map and a per-logical-qubit FIFO queue of pending gate IDs (derived directly from the circuit, independent of any router-internal DAG state), and checks, per emitted operation:

- every gate is popped from the head of *every* one of its operand qubits' queues before being marked executed, so gates cannot execute out of DAG order or twice;
- every `local_swap`/`local_cx` acts on a genuine intra-core coupling edge, at the physical positions the verifier's own simulated layout says the operand logical qubits occupy;
- every `teleport` uses the correct inter-core port pair for its two cores, both ports were free beforehand, and the source qubit was coupling-adjacent to the source port;
- every `cat_entangler`, `cat_cx`, and `cat_disentangler` triple observes port-adjacency and free-port preconditions, and no other operation touches either communication port while a cat section is open;
- the emitted `RouterStats` counters (EPR, teleport, telegate, packet, SWAP, local-CX, single counts) match the counts the verifier itself derives from the op stream, catching any bookkeeping drift between what is emitted and what is counted.

We ran the verifier against 60 randomized circuits (4–40 logical qubits, 20–600 gates, both architectures, seeds 0–59) and against every circuit in the evaluation suite (Section IV); all pass. Two further artefacts make these checks concrete: a *certified trace* (Figure 6) and a *fault-injection study* (Table XI).

A certified trace. Figure 6 replays the stream PORTER emits for the running example of Section III-G — fourteen operations exercising every op type — and lists, per operation, exactly what the verifier asserts before accepting it. The trace makes the three primitives' legality conditions operational: the TELEGATEPACKET is one *bracketed section* in which the two reserved ports are untouchable and every interleaved SWAP must carry packet provenance; the TELEDATA at op 11 must name the unique port pair of its core pair, find both ports free, start port-adjacent, and land exactly on the destination port; and every gate execution, local or remote, is popped from the head of each operand's pending-gate queue, so DAG order is enforced gate by gate rather than assumed from the router's own bookkeeping.

#	emitted operation		verifier asserts
0	`swap` $p_3 \leftrightarrow p_4$		coupling edge; operands sit where the replayed layout says
1	`cat_entangler` q_0; (p_5, p_{12}); 1 EPR		no section already open; *exact* port pair of (c_0, c_1); both ports free; q_0 adjacent to p_5; opens section, reserving p_5, p_{12}
2	`swap` $p_{10} \leftrightarrow p_9$	route q_1	tagged as packet routing; touches neither reserved port
3	`cat_cx` g_1 (p_{12}, p_9)		g_1 at the head of *both* q_0's and q_1's queues; control = section control; proxy is the target-side port p_{12}; target in c_1, adjacent to p_{12}
4	`swap` $p_{13} \leftrightarrow p_{14}$	route q_2	as op 2
5	`cat_cx` g_2 (p_{12}, p_{13})		as op 3
6	`cat_cx` g_3 (p_{12}, p_{15})		as op 3 (q_3 already port-adjacent)
7	`cat_disentangler`		a section is open; closes it, releasing the ports
8	`swap` $p_6 \leftrightarrow p_7$		coupling edge
9	`local_cx` g_5 (p_4, p_7)		queue heads; same core; coupling edge
10	`swap` $p_{13} \leftrightarrow p_{14}$		coupling edge
11	`teleport` q_5: $p_{13} \rightarrow p_5$; 1 EPR		cores adjacent; exact port pair (p_{12}, p_5); both ports free; q_5 adjacent to p_{12}; lands exactly on p_5
12	`swap` $p_4 \leftrightarrow p_5$		coupling edge
13	`local_cx` g_4 (p_3, p_4)		queue heads; coupling edge

Fig. 6. The full operation stream PORTER emits for the running example of Figure 5, as replayed by the verifier. Ops 1–7 execute inside the live cat section, where only packet-tagged SWAPs and 1Q gates that avoid the two reserved ports are legal. At end of stream the verifier further asserts: no cat section left open, all five gates executed exactly once, and the emitted statistics equal to the derived counts (2 EPR, 1 teleport, 1 packet, 0 single telegates, 6 SWAPs, 2 local CXs).

TABLE XI
FAULT-INJECTION VALIDATION OF THE VERIFIER (`VERIFY_FAULT_INJECTION.PY`): SINGLE-POINT CORRUPTIONS APPLIED TO OPERATION STREAMS THAT PASS VERIFICATION, EACH OF WHICH MUST BE REJECTED.

Injected corruption	Rejecting check
delete a `cat_disentangler`	section state machine
swap a teleport's two ports	exact inter-core port pair
zero a teleport's EPR cost	per-op EPR accounting
duplicate a gate-carrying op	every gate exactly once
delete a gate-carrying op	queue order / end-of-stream coverage
swap two ops sharing a qubit	per-qubit dependency order
retarget a SWAP to a non-coupled pair	coupling-edge legality
untag an in-section routing SWAP	cat-section isolation
retarget an in-section SWAP onto a port	reserved-port protection
increment `stats.epr` by one	stats–stream cross-check

Fault injection. On streams that always pass, a verifier that checks nothing is indistinguishable from one that checks everything, so we also certify that each check *fires*: `verify_fault_injection.py` takes passing op streams and applies the ten single-point corruptions of Table XI, one mutant per corruption, spanning misuse of all three communication primitives (wrong ports, free-port violations, broken cat-section bracketing), gate-coverage and dependency-order violations, illegal SWAPs, and statistics drift. Across the running example (which admits all ten corruption classes) and thirty fuzzed circuits on both architectures, all 276 applicable mutants are rejected, each by the intended check; the same harness runs on any benchmark QASM file.

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